

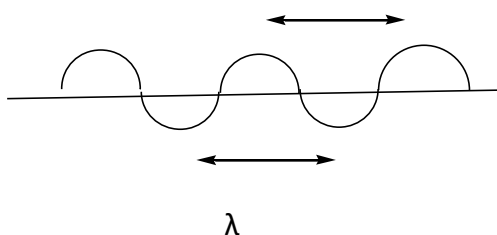
SPECTROSCOPY

Spectroscopy is an important physical method for the study of molecular structure of both organic as well as inorganic compounds. It is the study of interaction of electromagnetic radiation with matter. It involves the analysis of electromagnetic radiation emitted, absorbed or scattered by atoms or molecules.

Each type of electromagnetic radiation such as X-rays, UV rays, radio waves, visible light, etc., has a dual character. That is, they have both the properties of a wave as well as a particle. Electromagnetic radiations can be described as a wave occurring simultaneously in electrical and magnetic fields or as oscillating electric and magnetic fields perpendicular to each other. They can also be described as consisting of particles called quanta or photons. When electromagnetic radiation falls on matter, it interacts with atoms or molecules constituting the matter. This interaction results in the absorption of radiation.

Characteristics of electromagnetic radiations

1. Wavelength: The distance between consecutive crests (or troughs) of a wave is



called wavelength. The wavelength is expressed either in meters (m), millimeters (10^{-3}m), micrometers ($1\mu\text{m}=10^{-6}\text{m}$) or nanometer ($1\text{nm}=10^{-9}\text{m}$)

2. Frequency: The frequency of radiation, (ν), is the number of waves that pass a fixed point in unit time. It is expressed in reciprocal seconds (s^{-1}) or Hertz (Hz). λ and ν are related by the expression $\nu = c/\lambda$ where c = velocity of light = $2.998 \times 10^{10} \text{ cm sec}^{-1}$; and λ = wavelength in cm. thus ν and λ are inversely proportional.

3. Wavenumber ($\bar{\nu}$): It is the reciprocal of wavelength. i.e., $\bar{\nu} = 1/\lambda$. It is expressed in reciprocal cm (cm^{-1}).

4. Energy: The energy of a quantum or photon of electromagnetic radiation is related to the wavelength and frequency by the equation $E = h\nu = hc/\lambda$ where E = energy of radiation in Joule; h = Planck's constant = 6.626×10^{-34} ; ν = frequency in Hertz and c = velocity of light = $2.998 \times 10^8 \text{ m/s}$.

Thus energy of radiation is directly proportional to frequency and inversely proportional to wavelength. i.e., Energy is higher when frequency is higher and wavelength is smaller.

Electromagnetic spectrum

Electromagnetic spectrum is an ordered arrangement of electromagnetic radiations according to the wavelength. The wavelength of visible region ranges from 400 nm. to 750nm. The visible region, however, is a very small part of the entire electromagnetic

spectrum. Wavelength slightly shorter than those of the visible region falls into the UV region, while slightly higher wave length fall into the infra red region.

Regions of electromagnetic spectrum

<u>Spectral region</u>	<u>Wavelength</u>
Gamma rays	0.0001-0.01nm.
X-rays	0.01-2nm.
Ultra violet (UV)	200-400nm.
Visible light	400-750nm
Infra red (IR)	1.5-25 μ m
Microwave	1mm-30cm
Radiowave	10-2000m

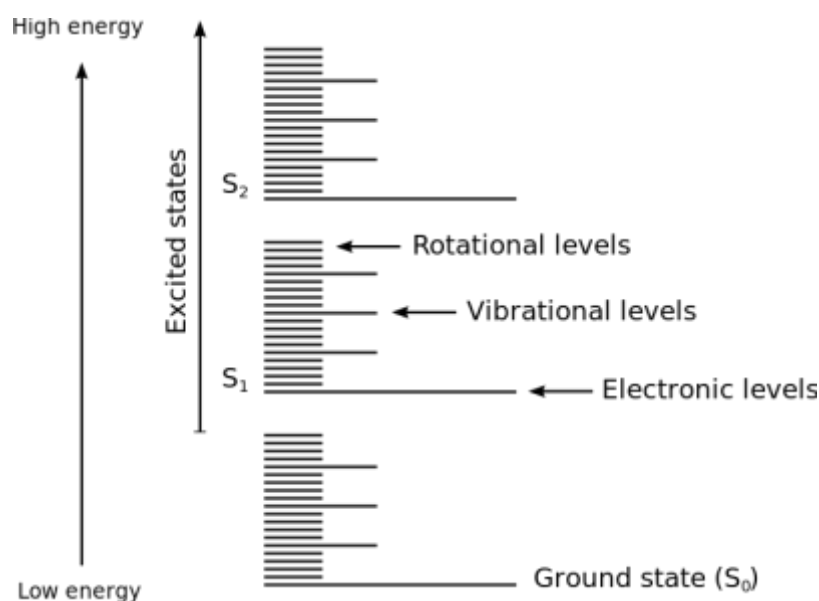
Absorption of electromagnetic radiation by molecules

When an electromagnetic radiation falls on matter it interacts with atoms/molecules constituting the matter. This interaction results in absorption of a portion of the radiation by the molecules. The amount of absorption depends on the wavelength of radiation and the structure of the compound. As a result of this absorption of light energy, the molecules undergo a transition from the normal ground state to an excited state of higher energy. Thus absorption of radiation leads to an increase in the total energy of the molecules.

Energy levels in the molecules

The total energy of a molecule is the sum of four different types of energies: translational energy, rotational energy, vibrational energy and electronic energy. Translational energy is concerned with the overall movement of molecules along three axes. Rotational energy is associated with spinning or rotation of the molecule about the axis passing through the centre of gravity. Vibrational energy is associated with the vibration of the molecule and, electronic energy is the energy associated with the distribution of electrons in various electronic energy states of the molecules. During interaction of radiation with a molecule, radiation is absorbed causing a change in rotational, vibrational or electronic energy levels of the molecules. Rotational, vibrational and electronic energies of a molecule are quantized, i.e., the molecules can exist only in distinct states which correspond to discrete or definite energy content. Molecules absorb radiations in discrete “packets” of $\Delta E = h\nu$, Which are also called quanta of energy. Absorption of radiation occurs only when radiation supplying exactly the right “packet” of energy sufficient for excitation to the next higher energy level.

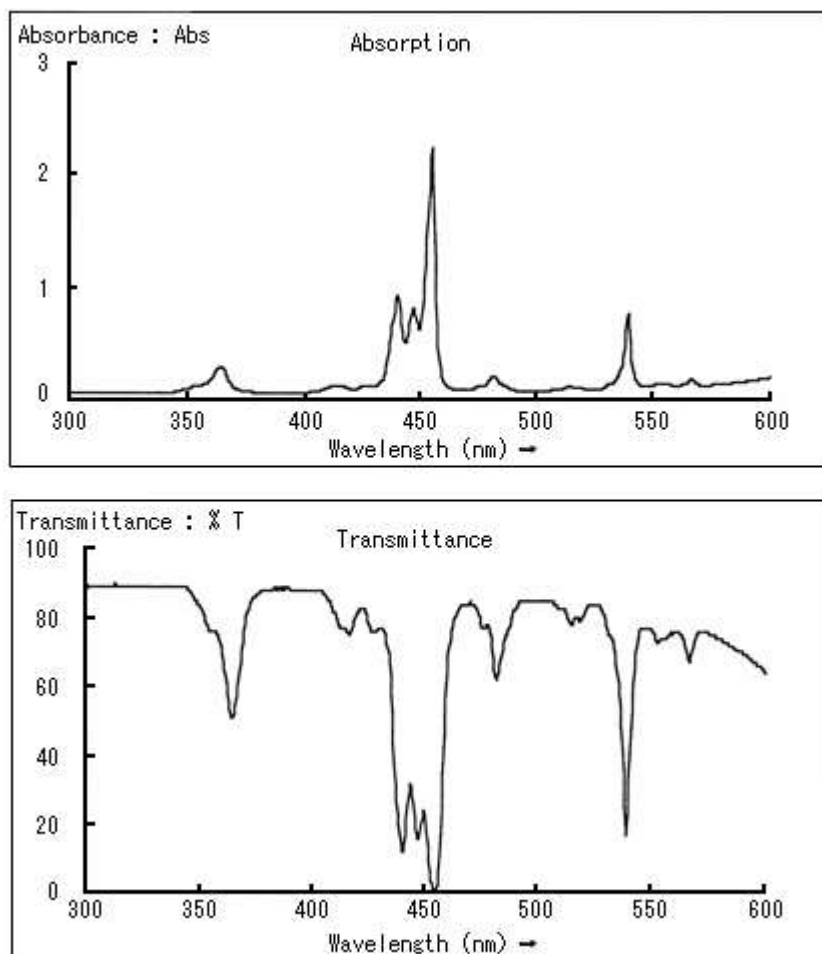
The electronic energy levels are designated by the quantum number n . n has integral values starting from 1 (i.e., $n=1,2,3, \dots$). Within each electronic energy level, there are a number of vibrational energy states designed by the quantum number V . V has integral values starting from zero (i.e., $V=0,1,2,3,\dots$). Within each vibrational states there are rotational energy levels which are designated by the quantum number J . J has integral values starting from zero(i.e., $J=0,1,2,3,\dots$)



The difference between successive electronic energy levels is of the order of 1000 to 10000 KJ/mole. For vibrational levels it is of the order of 1 to 100 KJ/mole and for rotational levels, it is of the order of 0.01 to 1 KJ/mole.

Absorption spectroscopy is of different types according to the type of radiations absorbed. In a given interaction with a particular type of radiation, the molecule may be absorbing some of the wavelengths of the radiations while the rest is transmitted. The absorption spectrum is recorded as a plot of either absorbance (A) or percentage transmittance (%T) versus wavelength or frequency or wave number.

Transmittance (T) = $\frac{I}{I_0}$ where I_0 = Intensity of incident light and I = Intensity of transmitted light. Similarly, absorbance, $(A) = \log_{10} \frac{1}{T} = \log_{10} \frac{I_0}{I}$



Law of absorbance of radiation

Beer- Lambert's law

The Beer-Lambert's law states that the proportion of light absorbed by a solute in a transparent solvent is independent of the intensity of the incident light and is proportional to the number of absorbing molecules in the light path. The Beer- Lambert's law can be expressed mathematically as

$$\log \frac{I_0}{I} = A = \epsilon cl$$

Where I_0 = Intensity of incident light,

I = Intensity of transmitted light,

ϵ = a constant called molar extinction coefficient or molar absorptivity,

c = Concentration of solute in moles/litre,

l = Cell (path) length in cm. and

A = Absorbance

when $c = 1$ mole/litre and $l = 1$ cm.

$$\log \frac{I_0}{I} = A = \epsilon$$

Thus, ϵ is the absorbance of a solution at a concentration of 1mole/ litre in a 1cm.cell.

Limitations of Beer- Lambert's law

Beer- Lambert's law is not obeyed when the radiation used is not monochromatic (i.e., containing single wavelength) and when the concentration of solution is higher than 10^{-3} M.

Selection rules

A **selection rule** describes how the probability of transition from one level to another cannot be **zero**. It has two sub-pieces: a **gross selection rule** and a **specific selection rule**. A gross selection rule illustrates characteristic requirements for atoms or molecules to display a spectrum of a given kind, such as an IR spectroscopy or a microwave spectroscopy. The specific selection rule determines whether a transition between certain specified quantum levels may happen or not. They specify the possible transitions among quantum levels due to absorption or emission of electromagnetic radiation.

A transition that is permitted by a specific selection rule is classified as **allowed**. Transitions that are disallowed by a specific selection rule are called **forbidden**. Gross and specific selection rules are different for different types of transitions.

Electronic spectroscopy (UV- Visible Spectroscopy)

Electromagnetic radiation in the UV (200-400nm.) and visible (400-750nm.) regions have sufficient energy for the promotion of outer electrons in molecules from a lower electronic energy level into upper levels.

Selection Rules of electronic transitions

For electronic transitions, there are no defined gross selection rules.

Electronic transitions may be allowed or forbidden transitions, as reflected by appearance of an intense or weak band, and is governed by the following **specific** selection rules :

1. Spin selection rule: The transition is allowed if $\Delta S = 0$. That is, there should be no change in spin orientation or no spin inversion takes place during these transitions. Thus, $S \rightarrow S$, $T \rightarrow T$ are allowed, but $S \rightarrow T$, $T \rightarrow S$ are forbidden transitions.
2. Laporte (orbital) selection rule ($\Delta l = \pm 1$): If the molecule has a centre of symmetry, there must be a change in the parity (symmetry) of molecular orbitals for the transition to occur. $g \rightarrow u$ or $u \rightarrow g$ transitions are allowed while $g \rightarrow g$ or $u \rightarrow u$ transitions are forbidden where g stands for gerade – Molecular orbital with a center of symmetry and u stands for ungerade – molecular orbital without a center of symmetry. Thus, $s \rightarrow p$, $p \rightarrow d$ are allowed, but $s \rightarrow s$, $p \rightarrow p$ are forbidden. i.e., transitions within a given set of p- or d-orbitals (i.e. those which only involve a redistribution of electrons within a given subshell) are forbidden.
3. $h\nu = \Delta E$ transition energy (photon matches energy gap between ground and excited state)

(Center of symmetry or inversion center, abbreviated *i*. A molecule has a center of symmetry when, for any atom in the molecule, an identical atom exists diametrically opposite this center an equal distance from it. In other words, a molecule has a center of symmetry when the points (x,y,z) and $(-x,-y,-z)$ correspond to identical objects. For example, if there is an oxygen atom in some point (x,y,z) , then there is an oxygen atom in the point $(-x,-y,-z)$.. Examples are xenon tetrafluoride , and benzene (C_6H_6).

Electronic transitions:

Since the outermost electrons are those involved in chemical bonding the characteristic observations provide information of the type of bonds present in molecule. Three distinct types of valence electrons are involved in organic molecules.

1. σ - electrons: These are involved in Saturated bonds such as between carbon and hydrogen in alkanes.

2. π - electrons ; These electrons are involved in unsaturated hydrocarbons like alkenes, alkynes, aromatic compounds, etc.

3.n-electrons: These are the electrons which are not involved in the bonding between atoms in molecules. Examples are organic compounds containing nitrogen, oxygen, chlorine, etc.

When a molecule absorbs UV radiations changes are produced in electronic energy of the molecule due to transition of valence electrons in the molecule. These transitions consist of the excitation of an electron from an occupied molecular orbital like a nonbonding p- or bonding π - orbital to an unoccupied orbital of higher energy like an antibonding π^* or σ^* orbital. Thus the promotion of an electron from a π -bonding orbital to an antibonding π^* orbital is designated as $\pi \rightarrow \pi^*$ (pi to pi star)

The relative energies of the important orbital are illustrated:

σ^* Antibonding

π^* Antibonding

non bonding

π – bonding

σ – bonding

***Types of Transitions**

1. $\sigma \rightarrow \sigma^*$ transitions: As the amount of energy required for transition of electrons from a bonding σ orbital to an antibonding σ orbital is much more than that available by UV light, such transitions are uncommon in UV spectrum
2. $n \rightarrow \sigma^*$ transition: This transition occurs in saturated compounds with one hetero atom such as oxygen, nitrogen, halogen, etc., with unshared pair (lone pair) of electrons (n-electrons). These transitions require comparatively less energy than $\sigma \rightarrow \sigma^*$ transitions. Such transitions do occur by absorption of UV radiations. So, any compound that contain atoms like N, O, S or halogen will absorb UV radiation.
3. $\pi \rightarrow \pi^*$ transition: These transitions are available in compounds with unsaturated centres. Eg. alkenes, aromatic compounds, carbonyl compounds, etc. This transition requires less energy than $n \rightarrow \sigma^*$ transition. Hence such compounds absorb UV radiation.

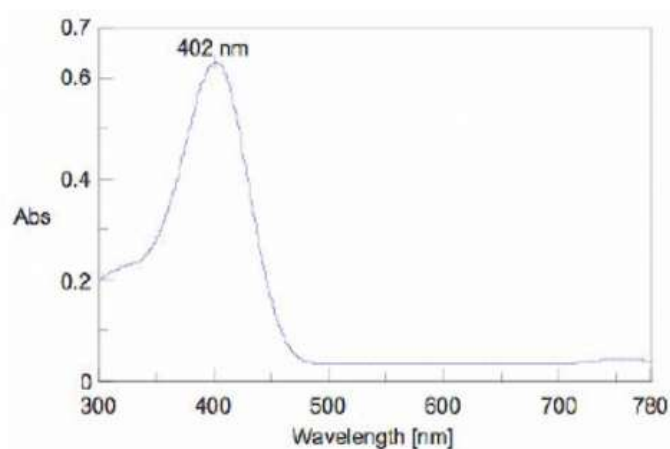
4. $n \rightarrow \pi^*$ transition: In this transition an electron of unshared electron pair on a hetero atom is excited to π^* antibonding orbital. This transition involves least amount of energy than all the transitions, and hence this transition gives rise to an absorption band at longer wavelength.

The energy values for different transitions are in the following order;

$$n \rightarrow \pi^* < \pi \rightarrow \pi^* < n \rightarrow \sigma^* < \sigma \rightarrow \sigma^*$$

Representation of UV- visible spectra

The UV spectrum is usually plotted as absorbance (i.e., $\log I_0 / I$ where I_0 is the intensity of the incident radiation I is the intensity of transmitted radiation) versus wavelength.



Chromophore and auxochrome

The unsaturated part or group of a molecule which is responsible for absorption of UV-visible radiation is called chromophore. Eg. Ethylene ($\pi \rightarrow \pi^*$ transition), carbonyl ($\pi \rightarrow \pi^*$ transition) and nitrile ($n \rightarrow \pi^*$ transition)

An auxochrome represents a saturated group which when attached to a chromophore changes both the intensity as well as wavelength of UV absorption. Eg. NH_2 , OH , Cl , etc.,. A shift of absorption to a longer wavelength is called a bathochromic shift (red shift). A shift of absorption to a shorter wave length is called a hypsochromic shift (blue shift)

Applications of UV-Visible spectroscopy

1. Qualitative analysis: UV spectroscopy is used to characterize aromatic compounds and conjugated alkenes. This technique is also used to differentiate conjugated dienes from non conjugated dienes and conjugated diene from conjugated triene. This is because as conjugation increases the maximum wavelength of absorption also increases. For example, ethylene ($\text{CH}_2=\text{CH}_2$) with one double bond absorbs UV radiation at 171nm., 1,3-butadiene($\text{CH}_2=\text{CHCH}=\text{CH}_2$) with a conjugated double bond absorbs at 217nm. and 1,3,5-hexatriene ($\text{CH}_2=\text{CHCH}=\text{CHCH}=\text{CH}_2$) absorbs at 256nm. Similarly vitamin A with a conjugated

system of five double bonds absorbs UV radiation at 325nm whereas lycopene with a conjugated system of eleven double bonds which is responsible for the red color of tomato absorbs radiation in the visible region. UV-visible spectroscopy is also used to distinguish between α, β – unsaturated ketones from β, γ – unsaturated ketones.

2. Quantitative analysis: UV spectroscopy is useful for the quantitative estimation of substances in solution. The absorptivity of the solution of the compound is measured for different concentrations, and plotted against concentration. The graph obtained is used for determining the unknown concentration of the solution.

3. Detection of impurities: this one of the best methods for detecting impurities in organic compounds since the different λ_{max} values for the impurities can be easily identified.

4. Chemical kinetics: UV spectroscopy is also used to study the kinetics of reactions. The kinetics of a reaction can be followed by measuring the absorbance since the concentration is proportional to absorbance.

5. Geometry of metal complexes: Comparing the wavelength of maximum absorption values of complexes of unknown geometry with those complexes with known geometry, the unknown geometry can be predicted.

6. Study of geometric isomerism: The trans isomer absorbs UV radiation at a longer wavelength and with a larger molar extinction coefficient than cis isomers.

Decay of the excited state

When a molecule absorbs light, an electron is promoted to a higher excited state. An electronically excited molecule can discard its excess energy by non-radiative decay or by radiative decay. The molecule can lose its energy non-radiatively by transferring its energy to another absorbing species in the vicinity or by collisions with other species in the medium. In radiative decay, an electron relaxes back into a lower energy orbital and in the process generates a photon. As a result, if the emitted radiation is in the visible region, the sample glows.

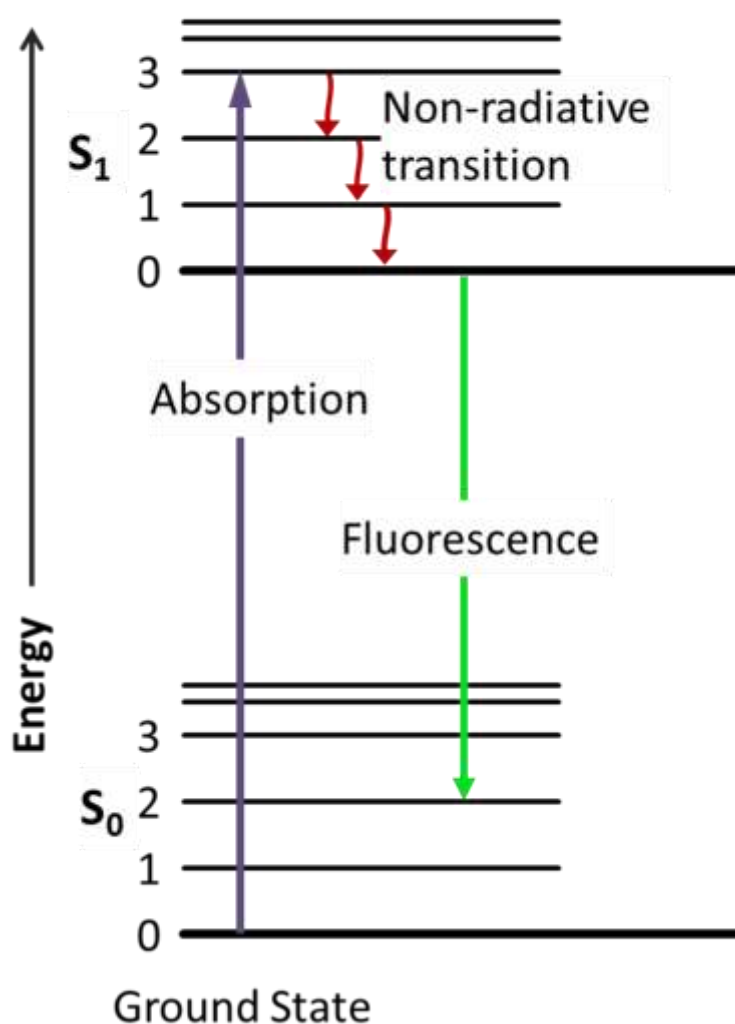
There are two principal modes of radiative decay, fluorescence and phosphorescence. In fluorescence, the spontaneously emitted radiation ceases very soon (within nanoseconds or milliseconds) after the exciting radiation is extinguished. In phosphorescence, the spontaneous emission may persist for long periods. In fluorescence, the emission is basically immediate and therefore generally only visible, if the light source is continuously on (such as UV lights); while phosphorescent material can store the absorbed light energy for some time and release light later, resulting in an afterglow that persists after the light has been switched off. Depending on the material, this afterglow can last anywhere from a few seconds to hours.

Fluorescence

Fluorescence refers to the spontaneous emission of radiation resulting from an electronic transition from the lowest vibrational level of the first excited electronic state to the one of the vibrational levels of the electronic ground state. The Jablonsky diagram given below

shows a schematic portrayal of molecular electronic and vibrational energy levels, and the sequence of steps involved in fluorescence.

At room temperature most molecules occupy the lowest vibrational level of the ground electronic state, and on absorption of light they are elevated to an excited electronic state.



Excitation can result in the molecule reaching any of the vibrational sub-levels associated with the excited electronic state. Having absorbed energy and reached one of the higher vibrational levels of an excited state, the molecule rapidly loses its excess of vibrational energy by collision and falls to the lowest vibrational level of the excited state. From this level, the molecule can return to any of the vibrational levels of the ground state, emitting its energy in the form of fluorescence. Fluorescence occurs at a lower frequency than that of

the incident radiation because the fluorescent radiation is emitted after some vibrational energy has been lost to the surroundings.

Beer- Lambert's law

1. At a definite wavelength, an absorber when placed in a cell of 1cm pathlength absorbs 20% of the incident light. If the absorptivity of the absorber at this wavelength is 2.0, find out its concentration. (Ans: 0.04846 mol/l)
2. A monochromatic radiation is incident on a solution of 0.05M concentration of an absorbing substance. The intensity of the radiation is reduced to one-fourth of the initial value after passing through 10cm length of the solution. Calculate the molar extinction coefficient of the substance. (Ans: 1.204 dm³mol⁻¹cm⁻¹)
3. In a cell, the aqueous solution of a substance of known concentration absorbs 10% of the incident light. What fraction of the incident light will be absorbed by the same solution in a cell 5 times as long? (Ans: 0.41)

Quantum yield

The fluorescence quantum yield (Φ , Φ_f) gives the efficiency of the fluorescence process. Quantum yield is defined as the efficiency of converting absorbed light into emitted light in the form of fluorescence. It is the ratio of photons emitted to photons absorbed. For highly fluoresce molecules, the quantum efficiency approaches to one. Molecules that do not fluoresce have quantum efficiencies that approach to zero.

Factors affecting Quantum yield

The bond of some organic molecules would rupture at 140 kcal/mol, which corresponds to 200-nm of radiation. For this reason, $\sigma^* \rightarrow \sigma$ transition in fluorescence are rarely observed. Instead, emissions from the less energetic transition will occur which are either $\pi^* \rightarrow \pi$ or $\pi^* \rightarrow n$ transition..

The most intense fluorescence is found in compounds containing aromatic group with low-energy $\pi^* \rightarrow \pi$ transitions. A few aliphatic, alicyclic carbonyl, and highly conjugated double-bond structures also exhibit fluorescence as well. The quantum efficiency increases as the number of rings and the degree of condensation increases.

Quantum efficiency of Fluorescence decreases with increasing temperature. As the temperature increases, the frequency of the collision increases which increases the probability of deactivation by non-radiative decay.

Solvents with lower viscosity have higher possibility of deactivation by non-radiative decay. Fluorescence of a molecule decreases when its solvent contains heavy atoms such as carbon tetrabromide and ethyl iodide, or when heavy atoms are substituted into the fluorescing compound.

Applications of fluorescence in medicine

At present, nearly all the diagnostic techniques and methods used for microorganism's diagnosis are not perfect and have some limitations. These techniques are time consuming, laborious and require many reagents. Fluorescence spectroscopy has great potential to become an excellent and perfect diagnostic technique for microorganisms. High sensitivity and specificity rate which makes Fluorescence spectroscopy an ideal diagnostic tool for medical microbiology field. Fluorescence emission spectra make it possible to distinguish between various bacterial species and also enable to classify the bacteria into genus, species and groups. Fluorescence spectroscopy may be a novel diagnostic tool to detect viruses. several diagnostic methods have been designed and tested to detect novel severe acute respiratory syndrome coronavirus - 2 (SARS-CoV-2), and one approved test kit utilises **fluorescence** based reverse transcript polymerase chain reaction (RT-PCR) [technique](#). The RT-PCR test works by detecting replicated nucleic acid copies from the SARS-CoV-2 virus through fluorescence of 5(6)-carboxyfluorescein

Also, viral infections of cells can be monitored by Fluorescence spectroscopy. Difference between fungal microorganisms like yeast can be made easily by use of spectroscopic fingerprinting.

Fluorescence spectroscopy has been applied successfully for the diagnosis of multisystem cancers with high sensitivity and specificity. Certain fluorescent materials are used as sensors in pathological labs for determination of glucose and monitoring of kidney functions. Fluorescent dyes like Ethidium bromide are used to quantify DNA and RNA.

Vibrational Spectroscopy (Infrared Spectroscopy)

It is one of the widely used tools for detection of functional groups in organic compounds and for comparison of compounds. The region of IR spectrum is 2-25 μm (wave number 4000-500 cm^{-1}). Organic compounds can absorb electromagnetic radiation in the IR region of the spectrum. IR radiation does not have sufficient energy to cause excitation of electrons. However, it causes atoms or group of atoms of organic compounds to vibrate faster about the covalent bond which connects them.

In their vibrations covalent bonds behave as if they were tiny springs connecting the atom. Atoms in a molecule do not remain at fixed positions with respect to each other, but actually vibrate back and forth about the inter atomic distance. This vibrational motion is quantized. At room temperature most of the molecules in a given sample are in the lowest vibrational state. However, on absorption of light of the appropriate energy the molecule becomes excited to the second vibrational level in which amplitude of vibration is greater.

Selection rules for vibrational spectroscopy

Gross selection rule

For a vibrational transition to occur as a result of absorption or emission of photon, the electric dipole moment of the molecule must change during the vibration. Thus, homonuclear diatomic molecules are said to be IR inactive because their dipole moments remain zero however long the bond. Heteronuclear diatomic molecules, which have a dipole moment that changes as the bond lengthens and contracts, are IR active. Thus, Carbon monoxide and iodine chloride absorb infrared light, but hydrogen (H_2), nitrogen (N_2), chlorine (Cl_2) and other symmetric diatomic molecules do not. But, for a molecule to be IR active it need not have a permanent dipole: the rule requires only a change in dipole moment, possibly from zero. Eg. CO_2

Specific selection rule

The specific selection rule for vibrational transition is that the change in vibrational quantum number, $\Delta v = \pm 1$

Types or modes of vibrations

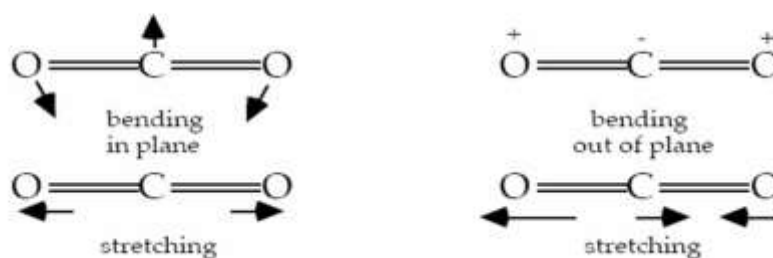
Atoms of molecule vibrate in a variety of ways. Thus, if atoms of a polyatomic non linear molecule are considered as balls connected by flexible springs, the laws of motion suggest that there will be $3n-6$ modes of vibrations. For example, H_2O molecule has ($3 \times 3 - 6 = 3$) vibrational freedom. They are symmetric stretching, asymmetric stretching and a bending vibration. In stretching the distance between two atoms increases or decreases, however, the atoms remain in the same bond axis.

symmetric stretch $\approx 3650\text{ cm}^{-1}$	asymmetric stretch $\approx 3760\text{ cm}^{-1}$	symmetric bend $\approx 1600\text{ cm}^{-1}$

These vibrations require higher energy and hence occur at higher frequency. In bending, the distance between two atoms remains constant, but the position of atoms may change with respect to the original bond axis. This type of vibration requires lower energy and occurs at lower frequency. The absorption frequencies for water molecule are 3652 (symmetric stretching) 3756 (asymmetric stretching) and 1595 (bending vibration)

For a linear polyatomic molecule there are $(3n-5)$ vibrational modes. Eg. CO_2 can have $(3 \times 3 - 5 = 4)$ modes of vibrations- symmetric stretching, asymmetric stretching and two bending vibrations in mutually perpendicular planes. Out of four vibrational modes, only three are IR active. The symmetric stretching does not involve the change of dipole moment and is not IR active.

For CO_2 , which is a linear molecule, there are $3(3) - 5 = 4$ fundamental vibrations:



The two bending modes are equivalent (degenerate). Therefore, only two absorption frequencies are observed in the case of CO_2 , at 2349 cm^{-1} for asymmetric stretching and 667 cm^{-1} for bending vibration.

Calculation of vibrational frequencies

According to Hooke's law, the frequency of vibration is directly proportional to the square root of the force constant of the bond. Force constant is a measure of the strength of a given bond. Again, the frequency is inversely proportional to the square root of the reduced mass, m_1m_2/m_1+m_2 of the molecule. The greater the mass, the lower the frequency of absorption. Similarly, stronger the bond the greater will be the frequency of absorption.

The vibrational frequency of a diatomic molecule, ν , is given by

$$\nu = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}$$

where k is the force constant and μ is the reduced mass. Reduced mass is given by

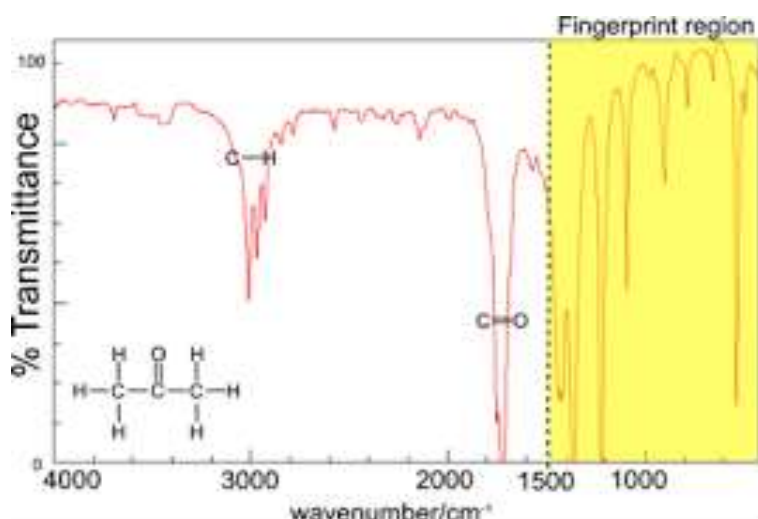
$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

Where m_1 = mass of atom 1 ; m_2 = mass of atom 2

Thus it can be predicted that C=C and C=O stretchings will have higher frequencies than C-C and C-O stretching respectively. Similarly, C-H and O-H stretching absorptions will be at higher frequencies than C-C and C-O stretchings.

An IR spectrum is obtained by plotting % transmittance ($100 \times \frac{I}{I_0}$ where I_0 is the intensity of incident radiation and I the intensity of transmitted radiation) against wave number.

The IR spectrum can be divided into two parts, the first between 4000 and 1600 cm^{-1} and the second between 1600- 500 cm^{-1} . The former region is called the functional group region since fundamental vibrations of most of the functional groups absorb in this region. The latter region is called the fingerprint region since compounds give a unique set of absorption bands in this region.



Some characteristic vibrations of IR spectrum

Bond	Group	Mode of vibration	wave number (cm^{-1})
C-H	Aliphatic	Stretching	2870-2960
C-H	Aromatic	„	3000-3100
C-H	CH=CH	„	3020-3080
C-H	CH=CH Cis	Bending	970
C-H	CH=CH Trans	„	690
C-C	C=C	Stretcing	1640-1675
C-C	C≡C	„	2100-2140

O-C	C=O	„	1600-1800
O-C	C-O	„	1250
O-H	free	„	3620

Applications of IR spectroscopy

1. Identification of functional groups: Most of the functional groups have characteristic IR absorption frequencies. For example, absorption range of C=O group in saturated ketones is $1700-1725\text{ cm}^{-1}$, OH group in alcohol is $3300-3500\text{ cm}^{-1}$, NH_2 group is $3200-3300\text{ cm}^{-1}$. Thus peak values in IR spectra help us to know the functional groups present in an unknown molecule.

2. Identification of unknown compounds: This is generally done by comparison of the finger print region of the IR spectrum of an unknown compound with that of a known compound.

3. Determination of purity: Presence of impurity can be detected since the impurities give rise to extra absorption bands.

4. Study of chemical kinetics: IR spectrum is very useful in following the progress of chemical reactions since most of the organic reactions involve change in functional groups. Practically, a small sample is withdrawn from time to time and spectrum is taken to determine to what extent the reaction has progressed.

5. Shape or symmetry of a molecule: IR spectroscopy is used to deduce the shape or symmetry of a molecule. e.g. NO_2 is proved to be a bent molecule since it gives three IR absorption peaks at 750 , 1323 and 1616 cm^{-1} in agreement with $(3n-6)$ formula applicable to tri atomic molecules.

6. Geometrical isomers can be distinguished by their IR spectra: Trans isomers are more symmetric than cis isomers. Hence trans isomer has little change in dipole moment in presence of IR radiation, and shows a simple spectrum than the cis isomer.

7. Presence of moisture in a sample: Since water has a characteristic absorption in the IR region, presence of moisture can be detected through IR spectroscopy.

8. In forensic analysis: Most of the evidence available at a crime spot is basically that of an organic compound. IR spectra can be used to detect human fluids, sweat, hair and toxic compounds.

Problem

The CO molecule absorbs IR frequency of 2140 cm^{-1} . Calculate the force constant of the chemical bond.

Rotational spectroscopy

Rotational spectroscopy involves transitions between the rotational energy levels of a gaseous molecule having a permanent dipole moment on the absorption of radiations falling in the spectral range of $1\text{--}100\text{ cm}^{-1}$ (microwave region). Free rotation is not possible for molecules in liquid or solid phase due to the presence of intermolecular forces.

Selection rules for Rotational spectroscopy

Gross selection rule

The molecule must have a permanent dipole moment. Thus H_2 (homonuclear diatomic), CH_4 (tetrahedral), and CO_2 (symmetric linear) molecules with no permanent dipole moment, do not show rotational spectra.

Specific selection rules

1. Transitions with $\Delta J = \pm 1$ are allowed; During the photon emission and absorption process, the rotational angular momentum J cannot change by more than one unit
2. $\Delta J = 0$ transitions are allowed when two different electronic or vibrational states are involved.

Rotational spectra of diatomic molecules:

A diatomic molecule (AB) can be assumed to behave as a rigid rotor, it means, internuclear bond distance does not change on rotation. The solution of Schrodinger equation for a rigid rotor gives

$E_J = \frac{\hbar^2}{2I} J(J+1)$ where I is the moment of inertia of molecule and is equal to μr_0^2 where μ is reduced mass of the molecule and r_0 is equilibrium bond length. For a diatomic molecule,

$$\mu = \frac{m_1 \times m_2}{m_1 + m_2}$$

The energy difference between two successive rotational energy levels is given by

$$\Delta E = E_{J+1} - E_J$$

$$\Delta E = \left[\frac{\hbar^2}{2I} (J+1)(J+2) \right] - \left[\frac{\hbar^2}{2I} J(J+1) \right] = \Delta E = \frac{\hbar^2}{I} (J+1) = \frac{h^2}{4\pi^2 I} (J+1)$$

Introducing the rotational constant B ,

$$\Delta E = 2hcB(J+1) \text{ where } B = \frac{h}{8\pi^2 Ic}$$

$$\text{i.e., } \frac{hc}{\lambda} = 2hcB(J+1) \text{ or}$$

$$\frac{1}{\lambda} = 2B(J+1)$$

Thus, if a spectral line corresponds to transition from J to J+1 level,

$$\frac{1}{\lambda} (\text{wave number}) = 2B(J+1)$$

The successive line will correspond to transition from J +1 to J+2

$$\frac{1}{\lambda'} = 2B(J+2)$$

Thus, the gap between two successive lines will be

$$\frac{1}{\lambda'} - \frac{1}{\lambda} = 2B(J+2) - 2B(J+1) = 2B$$

Hence the rotational spectrum of a diatomic molecule consists of lines at 2B, 4B, 6B, etc.

Applications of rotational spectroscopy

1. To determine the bond length
2. To differentiate between conformational isomers
3. Dipole moment of molecules can be obtained using rotational spectroscopy.
4. Relative abundance of isotopes can be obtained through rotational spectroscopy.

¹H NMR Spectroscopy (PMR Spectroscopy)

Nuclear Magnetic Resonance (NMR) involves the change in the spin state of a nucleus when the nucleus absorbs electromagnetic radiation in a strong magnetic field. The two types of NMR spectroscopy in common use are H¹ (proton –i.e., PMR) and C¹³ (carbon-13- i.e., CMR)

Principle of ¹H NMR spectroscopy

Some elements have isotopes with nuclei that behave as if they were spinning about an axis. The spinning of a charged particle generates a magnetic field. Isotopes with nuclear spin, therefore, have small magnetic fields, whose magnitude and direction can be described by a vector called magnetic moment. Only those nuclei display the NMR phenomenon for which the spin quantum number is greater than zero. The most important of such nuclei are ¹H and ¹³C, both of which have a spin quantum number of ½.

Many nuclei do not have spin. All even-even nuclei i.e., Those having an even number of protons and even number of neutrons, e.g., ¹²C, ¹⁶O belong to this class. In these nuclei, individual pairs of protons and neutrons have opposed spins and consequently the net spin of

the nucleus as a whole is zero. The numerical value of the spin number I is related to the mass number and atomic number as follows:

Mass number	Atomic number	Spin number
Odd	even or odd	$\frac{1}{2}$
even	even	0
even	odd	1

Thus, NMR is a phenomenon limited to compounds containing isotopes with either odd atomic number or mass number.

Since an electric charge is associated with an atomic nucleus, the spinning nucleus gives rise to a magnetic field. Thus, each nucleus for which $I > 0$ will have a characteristic magnetic moment. The proton (the nucleus of H atom) has a spin quantum number of $\frac{1}{2}$. If a magnetic nucleus is placed in a uniform magnetic field, the nucleus can have only a discrete set of orientations. The system is said to be quantized. The magnetic nucleus may assume any one of $(2I+1)$ orientations with respect to the applied magnetic field. Thus, a proton ($I=1/2$) will be able to assume only one of two possible orientations (called α -spin and β -spin) which correspond to energy levels of $\pm \mu H$ in an applied magnetic field Where H is the strength of the applied magnetic field and μ the magnetic moment which is a constant for a particular nucleus. The transition of a proton from one possible orientation to another (e.g. from α -spin state to β -spin state) may be effected by absorption of energy such that $\Delta E = h\nu = 2\mu H$ where ν is the frequency of electromagnetic radiation absorbed. In a common NMR spectrometer, a magnetic field of 14000 gauss (1.4 Tesla) is applied. At this field strength electromagnetic radiation of 60 MHz bring ^1H nucleus to flip from α -spin state to β -spin state. High resolution NMR spectrometers are available that operate at 100 MHz corresponding to a magnetic field of 23,486 gauss.

The NMR spectrometers are designed so that the radio frequency is kept constant, eg. At 60 MHz, and the magnetic field is varied. On tuning the magnetic field to the right strength, the protons in a molecule flip from a spin state to another by absorbing the radio frequency. A small electric current is generated due to the change in the spin state of proton. This current is displayed as a signal (or peak) in the spectrum.

Chemical shift

The field strength at which different protons of a molecule absorb radiations depends on their chemical environment. The electrons surrounding a nucleus, under the influence of a magnetic field, will circulate and in doing so generate their own magnetic field.. Since the induced field opposes the applied field the nucleus experiences a weaker magnetic field than that applied externally. This effect is called shielding. Since the effective magnetic field at the shielded proton weaker than the external field, the applied field must be made stronger for absorption of radiation to occur. Hence shielding is also called up field shift. The reverse effect in which the applied external field is shifted to lower fields is called downfield shift or de shielding. The shielding and de shielding effects are called chemical shifts since these shifts result from the circulation of electrons in chemical bonds. The chemical shift positions are frequently expressed in δ (delta) units which are defined as the differences, in ppm, from the signal for tetra methyl silane (TMS).

The degree of shielding or chemical shift depends on the electron density around the proton; the higher the electron density around the proton, the higher the shielding and the higher the field at which proton absorbs radiation. The shielding of a proton or chemical shift depends on the electronegativity of the atoms attached to it.

For example, consider the C-H and O-H bonds. O is more electronegative than C, therefore electron density around C - H proton is considerably greater than that around O-H

proton. Therefore the shielding effect is greater for C-H proton than O-H proton. Hence O-H proton will absorb at a lower field than C-H proton.

Measurement of chemical shift

Tetra methyl silane, $(\text{CH}_3)_4\text{Si}$, (TMS) is universally accepted as reference compound for representing chemical shift. It is chosen because it is a volatile liquid, b.p. 26.5°C , is inert to most reagents, soluble in most organic liquids. All the hydrogen atoms in TMS are equivalent and give an intense, sharp signal even at low concentrations. Furthermore, silicon is electropositive relative to carbon and tends to donate electron density to methyl group, thereby increasing their shielding. The relative high shielding of the protons in TMS causes it to absorb at fields higher than that for most protons commonly encountered in organic compounds.

The chemical shift positions are expressed in δ (delta) units which are defined as differences, in ppm, from the TMS signal. Chemical shifts are generally expressed in dimensionless units independent of applied frequency. This is done by dividing ν by the applied frequency and multiplied by 10^6 .

Thus a signal at 60 Hz ($\nu = 60\text{Hz}$) from TMS at an applied frequency of 60 MHz will be

$$\frac{60 \times 10^6}{60 \times 10^6}$$

Another standard scale is τ (Tau) scale. To convert from δ to τ , the chemical shift as measured on the δ - scale is subtracted from 10. i.e., $\tau = 10 - \delta$

Factors affecting chemical shift

1. Electronegativity: The electronegativity of the atom attached to the $-\text{CH}$ system decreases the electron density around the protons, thus reducing the shielding and increasing the chemical shift.

eg. CH_3 - 1.16; CH_3Br - 2.65; CH_3Cl - 3.10; CH_3F - 4.26

2. The number of electronegative substituents: The deshielding effects of electronegative substituents are cumulative.

eg. CHCl_3 - 7.3; CH_2Cl_2 - 5.3; CH_3Cl - 3.1

3. Distance from electronegative atom: eg. $\text{H}_3\text{C} - \text{CH}_2 - \text{CH}_2\text{Cl}$

1.0 1.4 3.4

4. Hydrogen bonding: Protons that are involved in hydrogen bonding change the chemical shift values. The more hydrogen bonding, more protons are deshielded, and the chemical shift value is higher.

5. Effect of solvent: the shielding or deshielding of the solvent can influence the chemical shift values of protons.

6. Magnetic anisotropy of π systems: Magnetic anisotropy is non uniform magnetic field. The anisotropy is due to the interaction of the electrons in the π systems with the applied magnetic field. It causes both shielding and deshielding.

Equivalent and non-equivalent protons

Protons which are in identical environment are called magnetically equivalent. If protons are magnetically equivalent, they have the same chemical shift, i.e., their absorptions occur at the same position in the NMR spectra. Eg. In $\text{CH}_3\text{CH}_2\text{Cl}$, the three methyl protons are equivalent. Similarly the two protons of CH_2Cl are also equivalent to each other. But the three protons of CH_3 group are not equivalent to two CH_2Cl protons, the latter being attached to the carbon carrying the electronegative chlorine atom. The three CH_3 protons have the same chemical shift and give a signal at the same position in the NMR spectrum. The two CH_2 protons have a greater chemical shift and give a separate signal. Thus the number of signals or peaks in a spectrum indicates the number of different kinds of hydrogen atoms in a molecule.

Spin-spin splitting

The splitting of signal lines (peaks) in an NMR spectrum into two or more components is called spin-spin splitting. Thus in the spectrum of $\text{CH}_3\text{CH}_2\text{Cl}$ the signal for CH_3 proton is split into a triplet (three lines) and the signal for CH_2 proton is split into a quartet (four lines). The number of peaks into which a proton signal is split equals one more than the number of nonequivalent neighboring protons, n . ($n+1$ rule). That is a doublet (d) results from one adjacent proton, a triplet (t) results from two adjacent protons a quartet from three adjacent protons, etc.,. Higher multiple peaks are recorded as multiplet(m). The spacing between the peaks in a multiplet resulting from spin-spin splitting is called coupling constant (J).

Spin-spin coupling is due to the effect of the spin arrangement of protons of one group on the protons of the adjacent group. The methyl (CH_3) group in $\text{CH}_3\text{CH}_2\text{Cl}$ is adjacent to the methylene (CH_2) group with two equivalent protons. For the two protons, there are three possible combinations of spin orientations as shown below:

$\uparrow\uparrow$	$\uparrow\downarrow$ and $\downarrow\uparrow$	$\downarrow\downarrow$
Both parallel	parallel and antiparallel	both antiparallel

If the effect of these spin arrangements of methylene protons can be transmitted to protons of methyl group, the magnetic fields of the protons of the methyl group will have three different values. The absorption of methyl proton would be then split into three peaks ($n+1$ rule).

Similarly, for the three protons of the methyl group, there are four different values of spin arrangements which affect the absorption of protons of methylene group.

$\uparrow\uparrow\uparrow$	$\uparrow\uparrow\downarrow$	$\downarrow\downarrow\uparrow$	$\downarrow\downarrow\downarrow$
	$\uparrow\downarrow\uparrow$	$\downarrow\uparrow\downarrow$	
	$\downarrow\uparrow\uparrow$	$\uparrow\downarrow\downarrow$	
All the 3	2 parallel	2 antiparallel	All the 3

parallel

1 Anti Parallel

1 parallel

Antiparallel

The CH₂ proton signal will, therefore, be split into four peaks (n+1 rule)

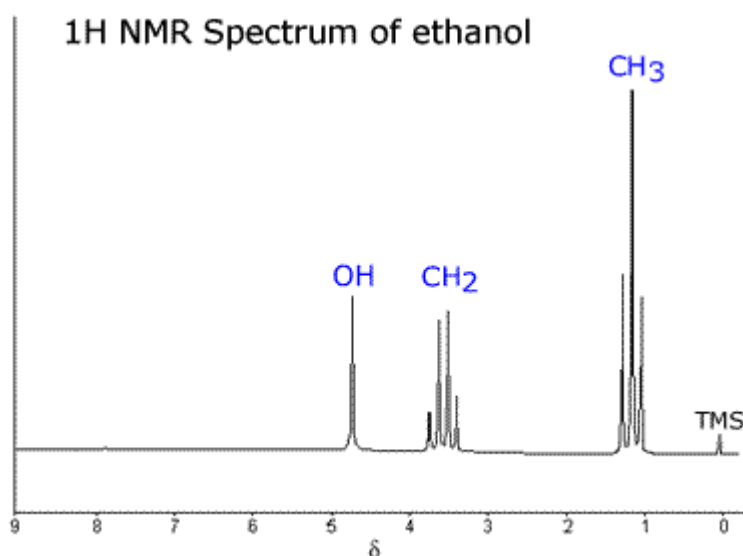
Applications of NMR spectra

I. Determination of structure by analysis of NMR spectra

NMR spectra can be used for determining the structure of unknown compounds. The important rules used for this are:

1. The number of main NMR signals is equal to the number of different kinds of hydrogen atoms present in the molecule.
2. The areas under each signal (peak) are in the same ratio as the number of hydrogen atoms causing each signal.
3. The number of peaks into which a signal is split equals one more than the number of nonequivalent neighboring protons.

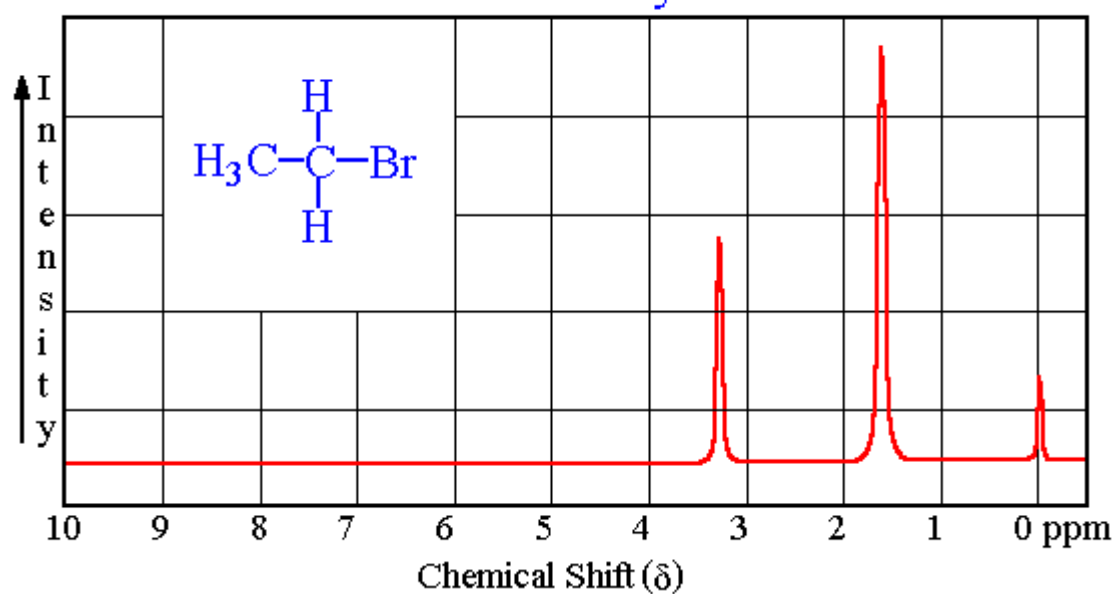
The above rules can be illustrated using the NMR spectrum of ethyl alcohol.



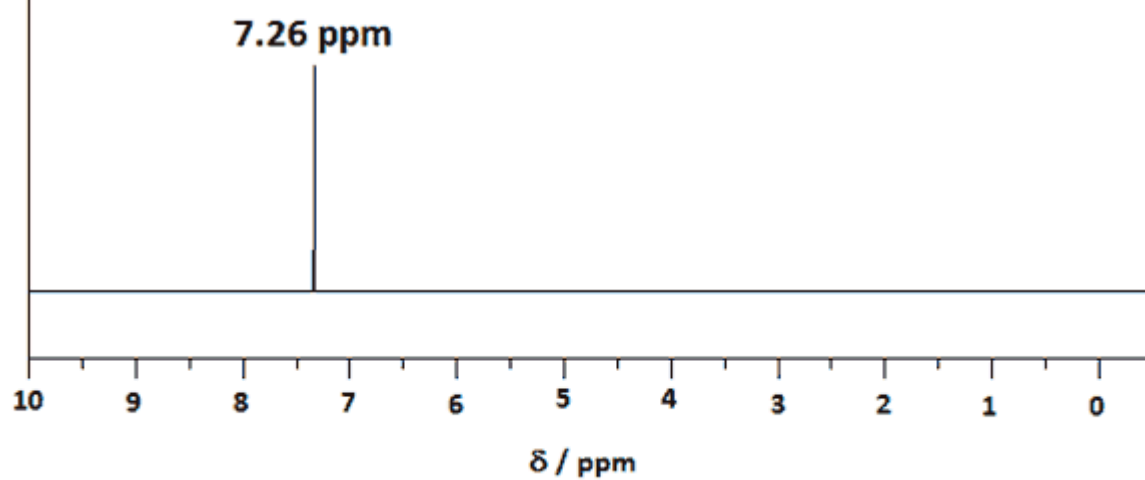
Ethanol has three different kinds of H atoms, and hence three major signals are obtained. The ratio of three peak areas is 1:2:3 and this corresponds to OH, CH₂ and CH₃. The peak for OH is a singlet since spin-spin splitting is not normally observed for a proton attached to oxygen or nitrogen atom. The peak for CH₂ group is a quartet since it has three neighboring protons. Similarly, the peak of CH₃ is a triplet since it has two neighboring nonequivalent protons.

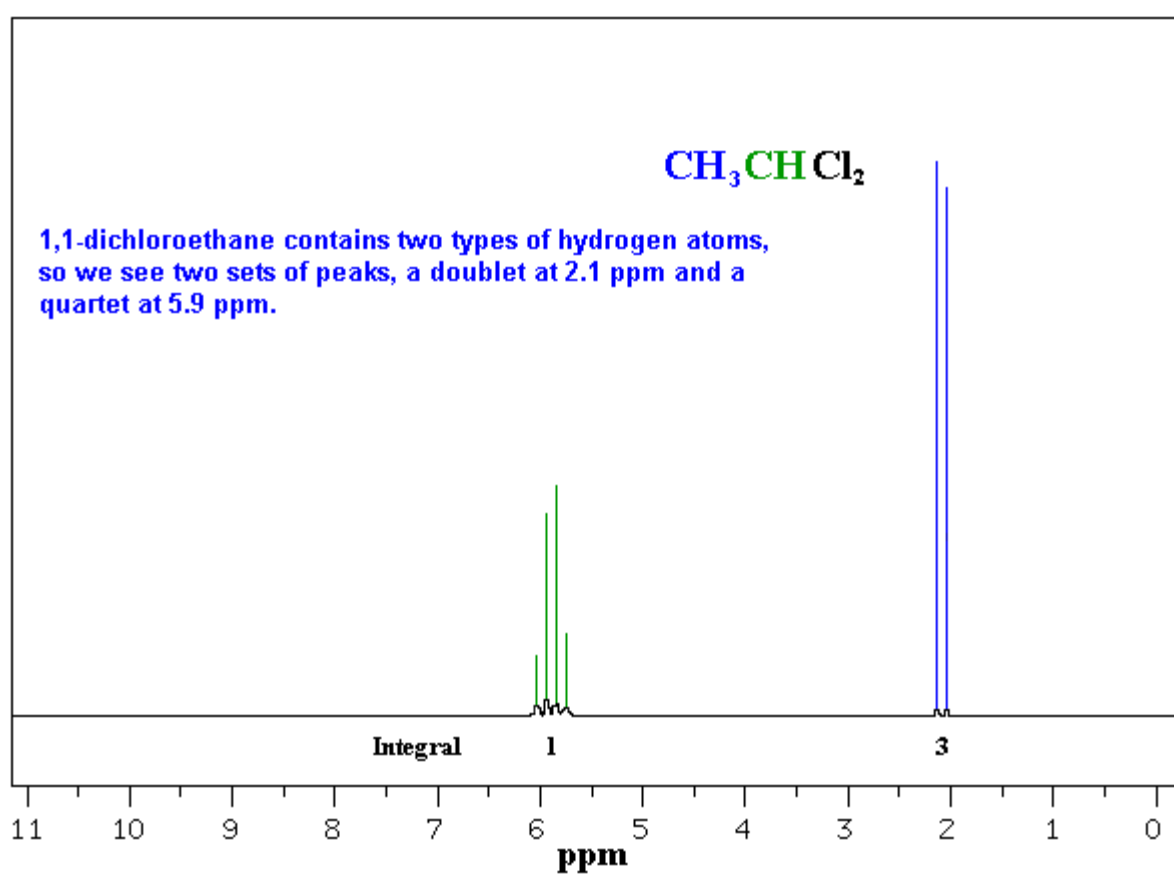
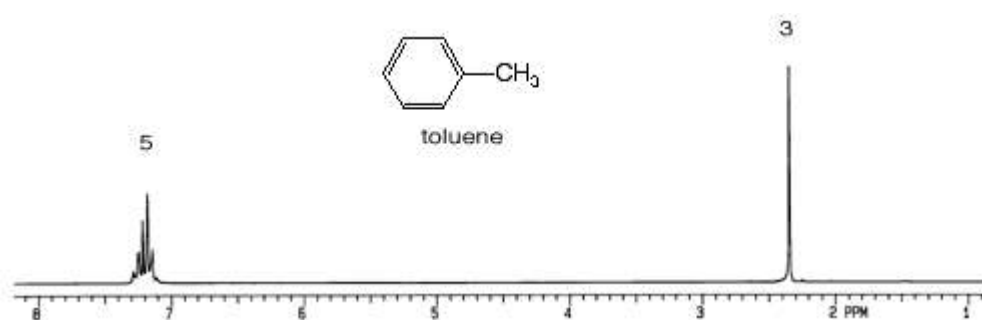
Another example is the spectrum of CH₃CH₂Br.

Proton NMR of Ethyl Bromide



^1H nmr of benzene





Magnetic resonance imaging (MRI)

MRI technology is another application of the principle of NMR spectroscopy. MRI is an extremely advanced technology widely used as a diagnostic method in medicine for producing image of tissues inside human body in order to diagnose and locate tumors and injuries in the body.

In MRI technology, the portion of the body to be diagnosed is subjected to a magnetic field. The magnetic field causes the protons (hydrogen nuclei) of the molecules present in the tissues inside the body to take up the specific alignment with respect to the magnetic field. Radio waves are then directed at the hydrogen nuclei. If the frequency of the radio wave is equal to that needed for excitation from the lower energy alignment of the nuclei to the higher energy alignment, the nuclei absorb radio waves. When the radio wave stimulation is stopped, the nuclei return to their original state and emit energy in the form of weak radio wave signals. These signals are characteristic of the tissues and can be converted to highly detailed cross-sectional images of the tissues with the aid of computer.

Though very expensive, MRI scanning has a high rate of success in detecting tumors, condition of chest, abdomen, head, spine and joints. MRI is safer since the radiations used are extremely low-energy radiations, namely radio waves.

Surface characterization techniques

In most materials, the surface composition differs from the bulk. The structure and properties of solid surfaces is crucial in many areas of science and technology such as metallurgy, microelectronics, catalysis, adhesion science, environment and corrosion. Therefore, diagnosis of surface structures is very important for materials science.

A complete characterization of the surface must provide information on the topography (roughness) of the surface, the geometrical (structural) arrangement of the surface atoms, the elementary composition, the lateral distribution of the elements, the in-depth distribution of the elements, etc. The properties of natural and fabricated material surfaces can be determined using characterization techniques such as X-ray diffraction and Raman scattering.

X-ray diffraction

One of the most important techniques for the determination of the structures of crystals is X-ray diffraction. X-rays are the electromagnetic radiations whose wavelength lies in the range of $0.02\text{\AA}$ to $100\text{\AA}$. Since this wavelength is smaller than that of visible rays, X-rays possess more energy than and higher penetration power.

A crystalline substance consists of atoms arranged in parallel rows separated by a specific distance called spacing. X-rays have wavelengths comparable to bond lengths in molecules and the spacing of atoms in crystals (about 100pm), so they are diffracted by

them. By analyzing the diffraction pattern, it is possible to draw up a detailed picture of the location of atoms. When an intense beam of X-rays strikes the crystal, it diffracts it depending upon the atomic arrangement in the crystal and orientation. X-ray diffraction patterns are produced when X-rays scattered from atoms in each set of lattice planes of a sample interfere constructively.

The theory behind the determination of crystal structure from its diffraction pattern is given by Bragg's equation, $n\lambda = 2d\sin\theta$ where θ = the incident angle with maximum diffraction intensity, λ = wavelength of the incident X-ray, d = the inner atomic spacing n , an integer, known as the order of reflection.

Applications of diffraction spectroscopy

It is the most widely used analytical technique for identification of the unknown crystalline sample. This technique is highly useful in studies of geology, material science and environmental studies as they characterize the crystalline materials and identify the fine-grained minerals like clays. It is also used to measure sample purity. XRD is also used to get the structure of proteins and their complexes.

Limitations of XRD

Only homogenous and single phase material can be identified properly by XRD. A mixture of minerals may not give accurate data. The sample must be powdered fine enough to get good results. Non-crystalline materials like glass cannot be analyzed using XRD technique.

Raman scattering

Raman spectroscopy is based on a special mechanism in which photons are scattered by molecules. In Raman scattering, about 1 in 10^7 of the incident photons collide with the molecules, give up some of its energy, and emerge with a lower energy. These scattered photons constitute the longer wavelength 'Stokes radiation' from the sample. Other incident photons may collect energy from the molecules that are already excited, and emerge as shorter wavelength 'anti Stokes radiation'. The component of radiation scattered without change of frequency is called Rayleigh radiation. A detector placed to collect energy at right angles to an incident beam will receive the scattered radiation.

If the molecule gains energy, say ΔE , the photon will be scattered with energy $h\nu - \Delta E$ (Stoke's radiation) and the equivalent radiation will have a frequency $\nu - \Delta E/h$. Conversely, if the molecule loses energy ΔE , the scattered frequency will be $\nu + \Delta E/h$ (anti Stoke's radiation)

Raman spectroscopy can be used to study rotational and vibrational transitions in molecules because the result of photon-molecule collision can be a change in the vibrational and rotational levels of molecules. Lasers are used as the source of the incident radiation in Raman spectroscopy because an intense beam increases the intensity of the scattered radiation.

Problems on Spectroscopy

1. How many proton signals would be expected in NMR spectra of each of the following compounds? Indicate also the multiplicity of various signals.

a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ b) CH_3OCH_3 c) $\text{CH}_3\text{OCH}_2\text{CH}_3$ d) $\text{CCl}_3\text{CHClCH}_2\text{Cl}$
e) $(\text{CH}_3)_2\text{CH-O-CH}(\text{CH}_3)_2$ f) $\text{CH}_3\text{CONHCH}_3$ g) $\text{CH}_3\text{CH}_2\text{COCH}_3$ h)
 $\text{CH}_3\text{CHBrCOOH}$

2. Give the structure of the compound with formula $\text{C}_9\text{H}_{10}\text{O}$ exhibiting at 1690cm^{-1} (IR) and ^1H NMR signals at $\delta 7.2(\text{m}, 5\text{H})$, $\delta 3.6(\text{s}, 2\text{H})$ and $\delta 2.0(\text{s}, 3\text{H})$.

3. From the following sets of proton NMR data, give a structure consistent with each of the following: a) $\text{C}_9\text{H}_{11}\text{Br}$ – $\delta 2.5(\text{m}, 2\text{H})$; $\delta 2.75(\text{t}, 2\text{H})$; $\delta 3.38(\text{t}, 2\text{H})$; $\delta 7.25(\text{s}, 5\text{H})$

b) $\text{C}_{10}\text{H}_{14}$ – $\delta 0.88(\text{s}, 9\text{H})$; $\delta 7.25(\text{s}, 5\text{H})$

B-Problems on Spectroscopy

1. How many proton signals would be expected in NMR spectra of each of the following compounds? Indicate also the multiplicity of various signals.

a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ b) CH_3OCH_3 c) $\text{CH}_3\text{OCH}_2\text{CH}_3$ d) $\text{CCl}_3\text{CHClCH}_2\text{Cl}$ e) $(\text{CH}_3)_2\text{CH-O-CH}(\text{CH}_3)_2$ f) $\text{CH}_3\text{CONHCH}_3$ g) $\text{CH}_3\text{CH}_2\text{COCH}_3$ h) $\text{CH}_3\text{CHBrCOOH}$

Ans: (a) 2 signals- t(6H), q(4H); (b) 1- s(6H); (c) 3- s(3H), q(2H), t(3H); (d) - 2-t(1H), d(2H); e - 2 - d(12H), m(2H); (f) - 3 - s(3H), s(1H), s(3H); (g) - 3 - t(3H), q(2H), s(3H); (h)- 3 - d(3H), q(1H), s(1H)

2. Give the structure of the compound with formula $\text{C}_9\text{H}_{10}\text{O}$ exhibiting at 1690cm^{-1} (IR) and ^1H NMR signals at $\delta 7.2(\text{m}, 5\text{H})$, $\delta 3.6(\text{s}, 2\text{H})$ and $\delta 2.0(\text{s}, 3\text{H})$.

Ans: IR band at 1690cm^{-1} indicates presence of -C=O group. NMR signal at $\delta 7.2(\text{m}, 5\text{H})$ indicates aromatic ring. Hence, probable structures are $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{CHO}$, $\text{C}_6\text{H}_5\text{COCH}_2\text{CH}_3$ & $\text{C}_6\text{H}_5\text{CH}_2\text{COCH}_3$.

Since the compound gives only 3 signals, the compound is not $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{CHO}$. As all the 3 signals are singlets, the compound is $\text{C}_6\text{H}_5\text{CH}_2\text{COCH}_3$. ($\text{C}_6\text{H}_5\text{COCH}_2\text{CH}_3$ should give s, q and t). Thus, signal at $\delta 7.2(\text{m}, 5\text{H})$ corresponds to protons of benzene ring, signal at $\delta 3.6(\text{s}, 2\text{H})$ corresponds to -CH_2 protons and signal at $\delta 2.0(\text{s}, 3\text{H})$ corresponds to -CH_3 protons.

3. From the following sets of proton NMR data, give a structure consistent with each of the following:

a) $\text{C}_9\text{H}_{11}\text{Br}$ - $\delta 2.5(\text{m}, 2\text{H})$; $\delta 2.75(\text{t}, 2\text{H})$; $\delta 3.38(\text{t}, 2\text{H})$; $\delta 7.25(\text{s}, 5\text{H})$

b) $\text{C}_{10}\text{H}_{14}$ - $\delta 0.88(\text{s}, 9\text{H})$; $\delta 7.25(\text{s}, 5\text{H})$

Ans: (a) NMR signal at $\delta 7.25(\text{s}, 5\text{H})$ indicates aromatic ring. Hence the probable structures are $\text{C}_6\text{H}_5\text{CHBrCH}_2\text{CH}_3$, $\text{C}_6\text{H}_5\text{CH}_2\text{CHBrCH}_3$ or $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$. Since the signals for the side chain are two triplets and one multiplet, the compound is not $\text{C}_6\text{H}_5\text{CH}_2\text{CHBrCH}_3$. As the peak area ratio of the signals of side chain is 2:2:2, the compound is $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$. Thus, the signal at $\delta 7.25(\text{s}, 5\text{H})$ corresponds to protons of benzene ring, signal at $\delta 3.38(\text{t}, 2\text{H})$ corresponds to $\text{-CH}_2\text{Br}$, signal at $\delta 2.75(\text{t}, 2\text{H})$ corresponds to middle -CH_2 and the signal at $\delta 2.5(\text{m}, 2\text{H})$ corresponds to -CH_2 attached to benzene ring.

(b) NMR signal at $\delta 7.25(\text{s}, 5\text{H})$ indicates aromatic ring. AS the compound is a hydrocarbon, the probable structures are $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$, $\text{C}_6\text{H}_5\text{CH}_2\text{CH}(\text{CH}_3)_2$ & $\text{C}_6\text{H}_5\text{C}(\text{CH}_3)_3$. Since there is only one signal for side chain, the compound is $\text{C}_6\text{H}_5\text{C}(\text{CH}_3)_3$. Thus, the signal at $\delta 7.25(\text{s}, 5\text{H})$ corresponds to protons of aromatic ring and that at $\delta 0.88(\text{s}, 9\text{H})$ corresponds to 3 methyl groups.